

Web-Based Stock Scoring Engine

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1 Introduction

Retail investors today have access to more financial data than ever before but access does not necessarily guarantee clarity. Investors can view earnings reports, valuation ratios, analyst headlines, and market data for companies across the S&P 500. However, the volume and inconsistency of this information can make decision-making difficult especially for beginners. This project, Web-Based Stock Scoring Engine, addresses that problem by converting selected financial and qualitative inputs into a ranked list of companies. Instead of requiring users to manually compare hundreds of data points across hundreds of firms, the system calculates a standardized score based on thirteen metrics and presents the results through a user-friendly interface. The goal is not to prove that these metrics guarantee strong investment returns but to build a transparent decision-support tool that helps users compare companies more efficiently.

This project matters because most investor-facing tools either over-simplify or over-complicate. Some investor-facing tools just show a few key stats and leave out a lot of important details. And others present extensive tables and charts but leave the synthesis entirely to the user. My approach is intentionally in the middle: automate the heavy computation then present both summary and drill-down views so users can quickly identify strong candidates and still inspect why each candidate ranked where it did. This design approach argues that financial interfaces should separate and organize information based on decision value, rather than only maximizing the amount of data displayed on screen. [5]. The project is being built as a web-based system with a Python backend and API layer. The current setup includes key ingestion workflows for S&P 500 companies and financial statement data with workflows built using the pandas library. In practical terms, the workflow consists of four stages: first, acquire a clean snapshot of the S&P 500 universe. Second, ingest raw fundamental and price data. Third, normalize these fields into metric components. And lastly aggregate them into a final score for ranking.

A major part of this Junior IS is going to consist of the integration of both quantitative and qualitative evidence into one ranking framework. Quantitative metrics (market cap, P/E Ratio, operating margin, short interest ...) are straightforward to compute from structured data. Qualitative metrics (leadership stability, competitive moat, news-driven catalysts ...) are harder because they depend on unstructured text and context. Rather than ignoring these qualitative signals, I plan to utilize a large language model (LLM) API to evaluate

qualitative signals and aim to convert these signals into comparable scores [8, 4]. This is important because many real investment decisions are not made from financial ratios alone.

Another contribution is interface design, even a statistically sound model can fail users if the output is hard to read. Lately, there's been a lot of focus in financial big-data visualization on making the results of models easier to understand. After a model prints out some results, the way we show that information should really highlight trends and summaries that users can actually act on. [3]. My system adopts that principle by prioritizing top-ranked summaries while preserving detailed metric breakdowns on demand. This project is not attempting to produce guaranteed price prediction or claim market-beating returns. The objective is to provide decision support rather than to make precise forecasts. Users should be able to see what data was used, what assumptions were made and how each component influenced rank position. [1].

Overall, this Junior IS presents the stock scoring engine as a link between the high amount of financial data and the quality of investor decisions. It offers a ranking system that's easy to follow, a scoring model that mixes different methods and a user-friendly approach to handling lots of market information.

2 Background

To understand the design, it is important to define the main concepts and constraints. At first, the plan was to rank all the stocks on the market, but because it was hard to find financial data for all of them, i had to narrow it down. So decided for the data to be based on the S&P 500, which is not just a random list of large U.S. companies. The index has clear rules from S&P Dow Jones Indices about which companies can be included and kept in it like liquidity and overall market structure.[10]. For a ranking web app, this matters because the results are only as good as the stocks being ranked. If the list includes stocks that should not be there, outdated ones or ones that are barely traded, the rankings can be misleading. That's why my project starts with a clear list of the actual constituents before any scoring happens.

Second, the project is built around working with data in tables. In Python, pandas library is the core tool because it makes it easy to work with labeled columns, handle missing values and merge data from different sources. [6].In practical terms,it lets our program combine company info, financial statement data, and price data into one clean table that's easy to compare.

Third, the web app looks at more than one thing. A stock can be cheap but not growing much or profitable but still risky financially. Since those factors are measured differently, I have to standardize them and give them weights before putting them together. That's why I use a multi-criteria approach, because it helps combine different strengths and weaknesses into one overall ranking.[1]. In my project, this means each metric adds to the final score without letting one single factor control the whole ranking.

Fourth, the system recognizes between hard and soft information. Hard information includes numeric and structured fields such as revenue, net income, debt, liquidity ... Soft information includes narrative and contextual signals such as management quality, strategic direction, and major news catalysts. Research I have read argues these categories should

be surfaced differently because users process them differently during decision-making [5]. My interface is built around that idea: hard metrics are there for direct comparison in the ranking table, while soft metrics are still shown but in a more supportive way, so they add context without making the main view feel crowded.

Fifth, working with qualitative data brings in NLP challenges. Financial news and text can be specific, fast-changing, and sometimes hard to interpret. Earlier systems like AZFin showed that pulling useful structure out of news can help make better market judgments.[8]. Later work found that character-level models handle specialized finance vocabulary and brand-new terms better than word-based models.[4]. For this project, these findings support my decision to use an LLM API to help process selected qualitative parts instead of relying on simple keyword counts.

Sixth, deployment matters because even if the ranking system works well in theory, it won't be very useful if updates are slow. [9]Since my current version is still early and not fully serverless yet, I'm not treating scalability as the main focus right now. But this research gives me a solid direction for how I could scale it later if the project ever grows beyond a class prototype.

Finally, UI/UX is not just about making the app look good. In a financial app, it actually affects how correctly people understand the data. Even if the numbers are right, a confusing design can still lead to bad conclusions. That's why things like readability, keeping the screen from feeling overwhelming and only showing extra details when needed are important for making the tool useful and trustworthy [7]. That perspective aligns with my design objective: a ranked table for quick understanding, plus deeper per-stock metric views for verification and user confidence. Together, these ideas make up the main focus of my Junior IS: choosing the right stock universe, building a reliable data workflow, combining different metrics into one ranking, using both quantitative and qualitative data, thinking about scalability, and designing the interface in a way that helps users make better decisions.

3 Related Works

The literature relevant to this project can be grouped into four themes: 1. financial information presentation and visualization, 2. quantitative scoring and ranking methods, 3. textual/AI approaches for qualitative data and 4. system architecture for real-time. Looking at these themes together helps show both what is already known and where my thesis adds something new. In terms of how financial information is presented, Ettredge, Richardson, and Scholz give a basic framework for how investment-focused websites should organize information[5]. Their distinction between hard and soft content remains directly useful for modern dashboards. The biggest strength of that work is how clear the idea is: it shows that just adding more numbers doesn't automatically help people make better decisions. The main limitation is that it's older, so it doesn't reflect how modern financial tools work today, for example, using AI to score qualitative info. My project builds on their idea by applying the hard vs. soft metric. More recent work by Dong, Huang, and Wang looks at financial big-data visualization from a machine learning perspective.[3]. What stands out most from their framework is the idea of looking at visualization before, during, and after modeling. The after-model stage matters the most for my project, since my system focuses

on presenting the final rankings in a way users can understand without feeling overloaded. Compared to Ettredge et al., Dong et al. pay more attention to large-scale data and model-based workflows. But their work is also broader and not really focused on stock-ranking dashboards. My project takes that post-model idea and applies it more directly to a ranking UI built around the S&P 500.

For quantitative scoring and ranking, MCDM literature gives the strongest method behind my approach. Behzadian et al.'s survey of VIKOR methods explains how different criteria that may conflict with each other can be adjusted and combined into one balanced ranking.[1]. This connects directly to the main challenge in my 13-metric model, which is combining different types of indicators without letting one metric overpower the rest just because it uses a bigger scale. The main point is not that I need to copy one specific algorithm, but that I need solid adjusting and weighting to make the final score fair and reliable. One thing the VIKOR literature does not really focus on is how to apply these ideas in a web-based financial dashboard that also includes qualitative data. My project helps fill that gap.

The bibliography also includes more recent work on improving UI/UX in financial software at the application level. [7]. This kind of work supports the idea that interface quality affects more than just how users feel about the app, it can also affect how they understand the results and the decisions they make. When combined with the older financial web-design model, it helps make a clear point. One limitation is that a lot of UX research stays pretty general and does not explain exactly how interface design should connect to the scoring system underneath. My project adds to that by linking UI choices directly to the ranking model, like showing an overall rank while also giving a clear breakdown of the individual metrics behind it.

For qualitative and text-based market signals, Schumaker and Chen's AZFin system is an important early example.[8]. It demonstrates that unstructured financial news can be transformed into predictive features through systematic text analysis. For my Junior IS, the main point is that qualitative signals don't have to stay as just opinions, they can be turned into something the system can measure and use. At the same time, AZFin comes from an older era of NLP, so it had tighter limits on how flexible the models were and what kinds of language they could handle compared to today's transformer-based systems. Building on that idea, the character-level neural approach by dos Santos Pinheiro and Dras goes a step further, showing that these models can handle finance-specific wording and even unfamiliar text patterns better than more word-based approaches. [4]. Still, neither paper fully solves the deployment and interpretability issues that come up in real dashboards. My project borrows from both to support AI-assisted soft metrics, while still keeping things transparent by showing the component scores behind the final ranking.

On system architecture, Solaimalai's serverless analytics paper shows a practical way to scale stock data collection and processing when demand goes up. [9]. This is valuable for the long-term roadmap of my project, which may need frequent updates across hundreds of symbols. The clear advantage of this literature is operational focus: latency, concurrency, and cost behavior are treated as first-class design concerns. The current limitation for my thesis stage is implementation scope. My prototype is currently pipeline- and API-centered rather than fully serverless. Nevertheless, the paper provides a credible architecture direction for future extensions. Finally, the S&P methodology document and pandas foundation paper

play a different but essential role. [10, 6]. The main strength of this work is that it focuses on the practical side of the system, like speed, handling multiple requests, and cost. For my project, the limitation is that my current version is still a prototype, so it is more focused on the system and API than on being fully serverless. Still, the paper gives me a solid direction for how the system could be built out later. Across all of these themes, one clear pattern shows up. A lot of existing research looks at important parts of the problem, but usually one piece at a time. Some focus on interface design without modern AI, some focus on ranking methods without building a full product, some look at NLP without explaining how results should be shown clearly in a dashboard, and others talk about system architecture without combining hard and soft scoring. The gap is a system that brings all of this together in a way that is solid technically, easy for users to understand, and realistic to build at the S&P 500 level. That is where my thesis fits in. It combines reliable data collection, multi-criteria scoring, qualitative signal analysis, and decision-focused visualization into one web-based ranking system.

4 Methodology

This project is about building a stock ranking system for companies in the S&P 500 that is easy to understand. The idea is to rank companies using both financial data and qualitative information, such as CEO-related context, while also making it clear to the user why a company ended up where it did in the ranking. I decided to work only on the S&P 500 because it gives a more reliable and realistic group of companies to work with. These firms are large, well-established, and usually have better reporting transparency and liquidity than smaller public companies [10]. This project starts by collecting the list of S&P 500 companies, then pulls financial and market data for each one (revenue, net income, debt, operating income, and related indicators). After that, the data is cleaned and organized using pandas so missing values or inconsistent entries do not affect the results [6]. Once that is done, the system scores them across thirteen different metrics and adds the scores together to create the final ranking. On top of that, the project also includes qualitative signals using LLM-assisted analysis. In the end, the goal is to produce a working web-based stock scoring engine where users can see the rankings, explore the scores, and understand the reasoning behind them.

This project does not rely only on financial ratios. It also includes qualitative information so the rankings provide a more complete picture of a company. Financial statements can show how a company has performed in the past, but they often miss factors such as leadership strength or whether the company has a clear plan for the future. To address that gap, the system uses an LLM to analyze less structured information, including company context and executive leadership profiles. For example, it can evaluate how a CEO is viewed and whether that leadership appears stable or risky. The LLM reads selected text through an API, scores it in areas such as market sentiment, strategic clarity, and management quality, and then converts those ratings into numeric components used in ranking [8, 4]. On the financial side, the metrics also need to be standardized because measures such as operating margin and short interest are expressed on different scales. Without a standardized scoring method, one metric could dominate the final score simply because its values are larger. To prevent that,

each metric is scored using predefined thresholds before final scoring, and missing values are removed. After that, both financial and qualitative scores are combined into the final score and shown in a web-based interface that allows users to compare companies and understand how the rankings were determined.

4.1 System Scope and Research Goal

The main goal of this project is to build a web-app that helps retail investors rather than a system that predicts stock prices or guarantees returns. That distinction matters because a lot of financial tools try to do too much and in doing so they either mislead the users or become too complicated to be useful. My system is designed around a simpler goal: take a large group of companies, evaluate them across a set of metrics that I thought was meaningful, and present the results in a way that helps users make more informed decisions.

The system focuses specifically on S&P 500 companies for a practical reason. The original plan was to rank all publicly traded stocks, but that turned out to be unrealistic for two reasons. First, the total number of publicly traded companies is simply too large to work with at this scale. Second, financial data for smaller and less covered companies is often incomplete or inconsistent, making it very hard to compute the full set of thirteen metrics reliably. Smaller stocks tend to have gaps in their reported financials, limited analyst coverage, and less standardized reporting, which means any score produced for them would be based on partial information. Narrowing the scope to the S&P 500 made it possible to build a system where the data is actually available and consistent enough to produce meaningful rankings [10].

The research goal for this Junior IS can be stated clearly: build and demonstrate a working pipeline that ingests S&P 500 financial data, normalizes it across thirteen defined metrics, incorporates qualitative signals through LLM assistance and outputs a ranked list that is readable and inspectable by a non-expert user. The system does not need to beat the market to be successful. It needs to consistently produce rankings that reflect the underlying data and present those rankings in a way users can verify.

4.2 Data Collection and Preparation

The first step in this project is building a reliable list of S&P 500 companies to work with. To do this, the system scrapes the constituent list from Wikipedia, which maintains an up-to-date table of all S&P 500 tickers and company names. While Wikipedia is not the most authoritative source for financial data in general, it is widely used for this specific purpose because it tends to stay current with index changes. From that, the system gets a list of all the tickers and their sector classification which is then used to pull the financial data for each company.

Once the stock universe is established, the system pulls financial and market data using the SimFin API. SimFin provides structured income statement, balance sheet, and share price datasets that can be downloaded in bulk and joined by ticker. For each company in the S&P 500 list, the system extracts fields such as revenue, operating income, net income, debt, cash, share price, and simple valuation helpers, then stores the results locally as CSV files. Storing the data locally is important for two reasons. First, it avoids repeatedly calling

the API on every run. Additionally, it ensures that the pipeline is reproducible, allowing rankings to be regenerated from the same data snapshot without needing a live connection.

Data cleaning is handled with pandas and is one of the more important steps in the pipeline [6]. Financial data from APIs is not always clean. Some companies report revenue figures in millions while others are in thousands. Some fields are missing entirely, especially for companies that recently joined the index. The cleaning process standardizes column names, converts all values to a consistent numeric format and removes any rows where critical fields are null. If a company is missing data for more than a certain number of metrics, it is excluded from the final ranking rather than given a misleadingly low score just because of missing inputs. This is a deliberate design choice: it is better to show fewer companies with reliable scores than to include all 500 with some scores that are only partially computed.

4.3 Metric Design, Normalization, and Weighting

The scoring model is built around thirteen metrics that together try to capture different dimensions of a company's quality. The thirteen metrics fall into a few different categories. Some are growth-focused, like revenue growth and earnings per share growth. Some measure profitability, like operating margin and return on equity. Others look at valuation, like the price-to-earnings and price-to-book ratios. And the rest are more risk-related, like debt-to-equity ratio and short interest percentage. I did not try to include every possible metric, the goal was just to cover the most important dimensions without having too much overlap between them.

One of the core challenges with combining these metrics is that they are all measured on completely different scales. Revenue growth might be expressed as a percentage between 0 and 10 while market cap is measured in billions of dollars and short interest is in percentages. If you just added these up directly, the metric with the largest numerical range would dominate the final score even if it is not actually the most important factor. To prevent this, each metric gets a score based on set rules instead of using a more constant normalization method. For every metric, there are predefined cutoff rules that map a company's raw value into one of three discrete scores: 0, 5, or 10 points. A company that falls below the lower threshold gets 0, one that falls in the middle range gets 5 and the one that clears the upper threshold gets 10. For example, a company with a strong operating margin would score 10 on that component, a company with a mediocre margin would score 5, and one with a weak or negative margin would score 0.

Once every metric is scored this way, the component scores are added together into a single score. Since there are thirteen metrics and each can contribute up to 10 points, the maximum possible score is 130. Companies are then ranked from highest to lowest based on that total. This method is easier to understand and clearer than continuous scaling because the rules for scoring are straightforward. A user can look at a company's breakdown and immediately understand why it received the score it did on any given metric [1].

For some metrics, a lower value is better rather than a higher one for example Debt-to-equity ratio, a lower ratio generally means less financial risk. For these inverse metrics, the threshold rules are reversed which means a company with a low debt-to-equity ratio scores a 10 instead of a 0.

4.4 Qualitative Signal Scoring

Financial ratios are very important for ranking, but they do not capture everything that matters when evaluating a company. Leadership quality, competitive position and recent news can all affect a company in ways that do not show up in the financial statements. A company might have perfect financials but be in the middle of a CEO transition that introduces real uncertainty. Another example is that a stock might look expensive on a P/E basis but have a newly announced product line that changes the growth a lot. These are the kinds of signals that the qualitative scoring is here to capture and why qualitative data is important.

To handle qualitative information in a systematic way, the system uses an LLM API to evaluate selected text inputs about each company and convert them into numeric scores. The inputs include company profile summaries and information about current executive leadership. The LLM evaluates these inputs across six dimensions using the same 0, 5, or 10 point threshold system used for the quantitative metrics. The six dimensions are competitive moat, leadership quality, secular trend alignment, company culture and mission, talent quality, and recession resilience. For example, a company with a durable competitive advantage would score a 10 on moat while a company with no clear edge would score a 0. Similarly, a company with good leadership and a strong track record of long-term execution scores a 10 on leadership while the other one with governance issues scores a 0. These six qualitative scores get added directly to the seven quantitative scores which brings the maximum possible total to 130 points across all thirteen metrics [8, 4].

The decision to use an LLM for this rather than simple keyword counting or sentiment analysis highlights the limitations of previous methods. Traditional sentiment analysis tools often struggle with financial terminology because words can have different meanings based on context. More importantly, evaluating whether a company has strong leadership or a credible strategic direction requires reading for meaning rather than just counting positive and negative words [4]. An LLM handles this better because it interprets context, weighs nuance and returns structured evaluations that are usable in a scoring pipeline.

One important design constraint is transparency. Since the qualitative scores are generated by a model, there is a risk that users may perceive them as objective facts rather than model-assisted estimates. To address this, the interface clearly labels which components of the scores come from qualitative LLM analysis compared to structured financial data.

5 Implementation

The stock scoring engine is built as a web-based application with a Python backend and a browser-based frontend. Python was selected because it is effective for data handling and integrates well with financial libraries, APIs, and LLM workflows [6]. Most of the processing happens in the backend, where the system follows a pipeline from data collection to final score output. It starts by retrieving the list of S&P 500 companies (via Wikipedia scraping), then pulls the financial and market data required for the model from the SimFin API. After collection, the data is cleaned because companies do not always report values in the same format and some fields can be missing. Pandas is used to structure the tables and handle

missing values [6]. Once cleaned, the data is stored in a structured format such as CSV so the system does not repeatedly call the same APIs which improves runtime efficiency. The metrics are then scored using a combination of quantitative rules and Gemini/LLM-assisted qualitative evaluation. For each batch of companies, the system prepares selected structured company data and sends it through a structured Gemini prompt that asks the model to evaluate six qualitative dimensions: competitive moat, leadership quality, secular trend alignment, culture/mission, talent quality and recession resilience. The qualitative scores are converted into the same 0, 5 or 10 point scale used by the quantitative metrics which allows both results to be combined consistently. After both quantitative and qualitative scores are calculated, the system stores the component scores, computes the final total and sends the ranked results to the frontend. Users can then view the ranked list of companies and inspect the score breakdown for each company to understand why it received its position.

6 Results

This section is not meant to prove that the system can beat the stock market. The point is to show that the engine can actually do what it was designed to do. In this project, a good result means the system can take the list of S&P 500 companies, collect the needed data, calculate the scores, and present the rankings in a way that users can understand. The main output is a ranked table of companies based on their overall scores. That table shows the ticker, company name, overall score, and the breakdown of the thirteen metrics used in the model. This matters because users should be able to look beyond the final ranking and understand what pushed a company higher or lower. For example, one company might score well in profitability but poorly in qualitative areas, while another may be more balanced across the board, and those differences should be visible in the final breakdown.

6.1 Ranking Output Examples



Quantitative Ranking 10 / 503 shown

Soft metrics: 503 / 503 scored (cached)

RANK	TICKER	TOTAL (130)	SECTOR	REVENUE	P/E	DEBT
1	ANET	95 / 130	Information Technology	\$2.0B	242.7	\$0
2	GOOG	90 / 130	Communication Services	\$90.2B	117.3	\$10.9B
3	CDNS	90 / 130	Information Technology	\$1.2B	292.4	\$2.5B
4	MSFT	90 / 130	Information Technology	\$70.1B	113.6	\$42.9B
5	LLY	90 / 130	Health Care	\$12.7B	301.1	\$38.5B
6	NVDA	85 / 130	Information Technology	\$130.5B	66.9	\$8.5B
7	ISRG	85 / 130	Health Care	\$2.3B	243.9	\$0
8	NOW	85 / 130	Information Technology	\$3.1B	199.8	\$1.5B
9	VRTX	85 / 130	Health Care	\$11.0B	-213.9	\$0
10	ADBE	85 / 130	Information Technology	\$5.7B	58.1	\$6.2B

Show more

Figure 1: Prototype ranking table (top rows shown).

Figure 1 shows the top-ranked companies from a prototype run. The scoring output includes both hard financial metrics and soft qualitative metrics. This produces a more useful ranking because companies are no longer evaluated only by financial ratios but also scored on qualitative signals such as leadership quality, competitive moat, long-term trend alignment and recession resilience. The table shows that the system can process a large set of S&P 500 companies, calculate component scores and present a ranked output that can be inspected by the user.

6.2 Tables, Figures, and Metric Breakdowns

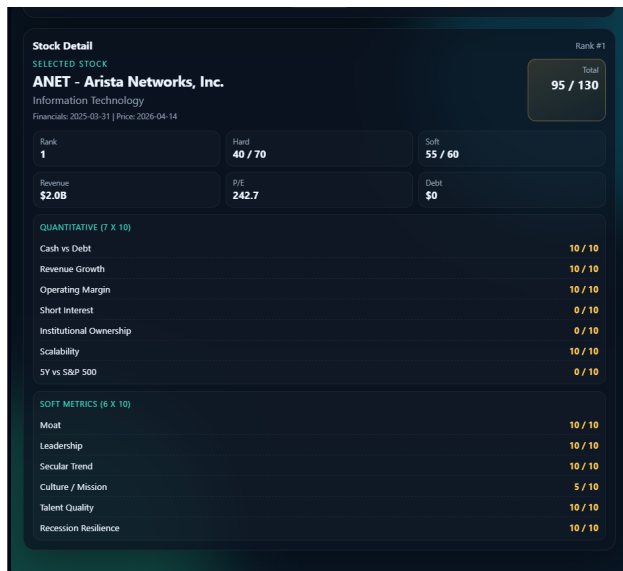


Figure 2: Stock detail scorecard output for ANET in the dashboard.

Figure 2 shows the stock detail view for ANET (Arista Networks) ,one of the companies scored by the current prototype .The detail page separates the final score into quantitative and qualitative components so users can see how each part contributes to the total ranking .The quantitative side includes metrics such as cash versus debt, revenue growth, operating margin, short interest, institutional ownership, scalability and S&P 500 comparison. The qualitative side includes LLM-assisted scores for moat, leadership, secular trend, culture/mission, talent quality and recession resilience . This breakdown supports the project’s transparency goal because the user is not only shown a final score but also the reasoning structure behind that score.

7 Conclusion

This project set out to build a web-based stock scoring engine that helps retail investors cut through the noise of financial data and evaluate S&P 500 companies more efficiently. The system ranks companies using a transparent 0/5/10 threshold scoring model across thirteen metrics: seven quantitative metrics and six qualitative metrics. It then presents those rankings through an interface that shows both a summary score and a full metric breakdown. The quantitative metrics evaluate financial and market-based factors while the qualitative component uses LLM-assisted analysis to score competitive moat, leadership quality, secular trend alignment, culture/mission, talent quality and recession resilience.

The main contribution of this project is bringing these pieces together in a single system. Most related work addresses one part of the problem at a time such as financial visualization, ranking methods, qualitative text analysis or system architecture. This project combines those areas into one working prototype while keeping the scoring logic visible to the user

rather than hiding it in the backend. That transparency is intentional because users should be able to understand why a company received a certain score instead of only seeing a final ranking.

There are still limitations. First, the thirteen metrics and their thresholds were selected manually so the rankings reflect a structured scoring framework rather than a statistically validated investment strategy. Second, the qualitative scores depend on the quality of the company information given to the LLM and the consistency of the model’s evaluation. These scores help include context that financial ratios may miss. Third, the system depends on external financial APIs for company and market data which creates a data freshness issue. The API does not return the most recent financial statements, market values, analyst data or company updates on the free version of the API. Because the free API tier may not provide the most recent financial statements some rankings may be based on delayed information. This matters because stock evaluations can change quickly after earnings reports. Future work should focus on validating the scoring model against historical stock performance, refining the metric thresholds, improving the qualitative prompts and improving data freshness if better API access becomes available. A paid financial data API could allow the system to refresh statements, price data and company metrics more frequently. The existing timestamp fields could then be used to show users exactly how recent each score is and to flag outdated records before rankings are calculated.

References

- [1] BEHZADIAN, M., OTAGHSARA, S. K., YAZDANI, M., AND IGNATIUS, J. A state-of-the-art survey of vikor methodology. *International Journal of Information Technology & Decision Making* 11, 01 (2012), 139–181.
- [2] CORMEN, T. H., LEISERSON, C. E., RIVEST, R. L., AND STEIN, C. *Introduction to Algorithms*, 3rd ed. MIT Press, Cambridge, MA, 2009.
- [3] DONG, X., HUANG, W., AND WANG, J. Financial big data visualization: A machine learning perspective. In *Proceedings of the 17th International Symposium on Visual Information Communication and Interaction (VINCI '24)* (New York, NY, USA, 2024), ACM. Accessed: 2026-01-27.
- [4] DOS SANTOS PINHEIRO, L., AND DRAS, M. Stock market prediction with deep learning: A character-based neural language model for event-based trading. In *Proceedings of the Australasian Language Technology Association Workshop 2017* (2017), pp. 6–15. Accessed: 2026-02-19.
- [5] ETTREDGE, M., RICHARDSON, V. J., AND SCHOLZ, S. A web site design model for financial information. *Communications of the ACM* 44, 11 (2001), 51–55. Accessed: 2026-01-27.
- [6] MCKINNEY, W. Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference* (2010), S. van der Walt and J. Millman, Eds., pp. 56–61. Accessed: 2026-01-27.

- [7] RUNSEWE, O., AKWAWA, L. A., FOLORUNSHO, S. O., AND OSUNDARE, O. S. Optimizing user interface and user experience in financial applications: A review of techniques and technologies. *World Journal of Advanced Research and Reviews* 23, 03 (2024), 934–942.
- [8] SCHUMAKER, R. P., AND CHEN, H. Textual analysis of stock market prediction using breaking financial news: The azfin text system. *ACM Transactions on Information Systems* 27, 2 (2009), 12:1–12:19. Accessed: 2026-01-27.
- [9] SOLAIMALAI, G. Serverless architecture for real-time stock market data analytics in cloud environments. *International Journal of Computer Applications* 186, 75 (2025), 9–15.
- [10] S&P DOW JONES INDICES. S&p u.s. indices methodology. Tech. rep., McGraw Hill Financial, February 2016. Accessed: 2026-01-27.